

Module 7 : Rerandomization

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Design of Experiments - Stat 140

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Rubin (2008a), Rubin recounts the following conversation with his advisor Bill Cochran:

Rubin: What if, in a randomized experiment, the chosen randomized allocation exhibited substantial imbalance on a prognostically important baseline covariate?

Cochran: Why didn't you block on that variable?

Rubin: Well, there were many baseline covariates, and the correct blocking wasn't obvious; and I was lazy at that time.

Cochran: This is a question that I once asked Fisher, and his reply was unequivocal:

Fisher (recreated via Cochran): Of course, if the experiment had not been started, I would rerandomize.

A similar conversation between Fisher and Savage, wherein Fisher advocates rerandomization when faced with an undesirable randomization, is documented in Savage [(1962), page 88].

- Hot topic formally introduced by Morgan and Rubin (2012).
- Sometimes you end up with a bad randomized allocation, i.e., there is a chance that imbalances exist in covariates between treatment groups.
- Provided a precise definition of imbalance has been specified in advance, unbalanced randomizations can be discarded.
- Advantages?

Procedure

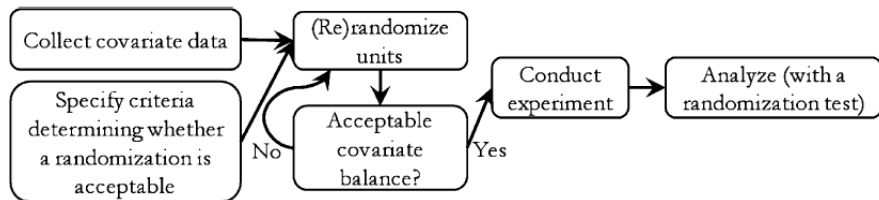


FIG. 1. *Procedure for implementing rerandomization.*

2.1. *Procedure.* The procedure for implementing rerandomization is depicted in Figure 1, and has the following steps:

- (1) Collect covariate data.
- (2) Specify a balance criterion determining when a randomization is acceptable.
- (3) Randomize the units to treatment groups.
- (4) Check the balance criterion; if the criterion is met, go to Step (5). Otherwise, return to Step (3).
- (5) Conduct the experiment using the final randomization obtained in Step (4).
- (6) Analyze the results using a randomization test, keeping only simulated randomizations that satisfy the balance criterion specified in Step (2).

- Let us assume that N units have been selected and SUTVA holds.
- Let W be the N -dimensional treatment assignment vector indicating the treatment group for each unit :

$$W_i = \begin{cases} 1 & \text{if unit } i \text{ is treated} \\ 0 & \text{if unit } i \text{ is control} \end{cases}$$

- Causal estimand : $\tau = \overline{Y}(1) - \overline{Y}(0)$

- Let Y_i^{obs} denote the vector of observed outcome values :

$$Y_i^{obs} = W_i^{obs} Y_i(W_i = 1) + (1 - W_i^{obs}) Y_i(W_i = 0)$$

- Estimator of τ :

$$\hat{\tau} = \overline{Y_T^{obs}} - \overline{Y_C^{obs}}$$

- $\overline{Y_T^{obs}} = \frac{1}{N_T} \sum_{i: W_i=1} Y_i^{obs}$
- $\overline{Y_C^{obs}} = \frac{1}{N_C} \sum_{i: W_i=0} Y_i^{obs}$

- Let us consider all the observed covariates for which balance between groups is desired.
- Let X be a $N \times K$ matrix representing K functions of covariates measured :
 - Original covariates
 - Any functions of original covariates (e.g., powers, logs, interactions).

Rerandomization

- The rerandomization criterion is based on a row-exchangeable scalar function of X and W :

$$\varphi(X, W) = \begin{cases} 1 & \text{if } W \text{ is an acceptable randomization} \\ 0 & \text{otherwise} \end{cases}$$

- $P(\varphi(X, W) = 1)$: proportion of acceptable randomizations.
- The choice of $P(\varphi(X, W) = 1)$ involves a trade-off between better balance and computational time :
 - Smaller values of $P(\varphi(X, W) = 1)$ ensure better balance, but they also imply a longer expected waiting time to obtain an acceptable randomization.

Rerandomization using Mahalanobis distance

- Affinely invariant scalar measure of multivariate covariate balance and takes the correlation between covariates into account.

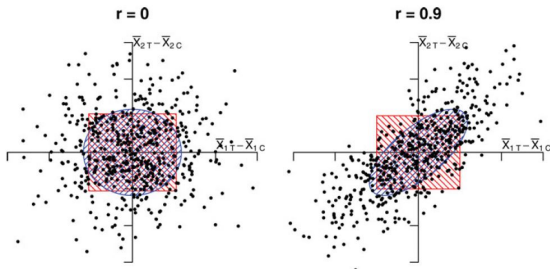


Figure 4. Points displayed are $\bar{X}_{j,T} - \bar{X}_{j,C}$ for two covariates, under orthogonality (left) and correlation (right), under pure randomization of units into two equally sized treatment groups. Acceptance regions are shown for two different rerandomization methods, both with $p_u = 0.5$. The squares are the acceptance regions restricting each covariate individually and the ellipses are the acceptance regions based on M .

Rerandomization using Mahalanobis distance

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Rerandomization using Mahalanobis distance

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$$\begin{aligned} M &= M(\bar{X}_T; \bar{X}_C) = (\bar{X}_T - \bar{X}_C)' [\text{cov}(\bar{X}_T - \bar{X}_C)]^{-1} (\bar{X}_T - \bar{X}_C) \\ &= \frac{1}{\frac{1}{N_T} + \frac{1}{N_C}} (\bar{X}_T - \bar{X}_C)' [\text{cov}(X)]^{-1} (\bar{X}_T - \bar{X}_C) \end{aligned}$$

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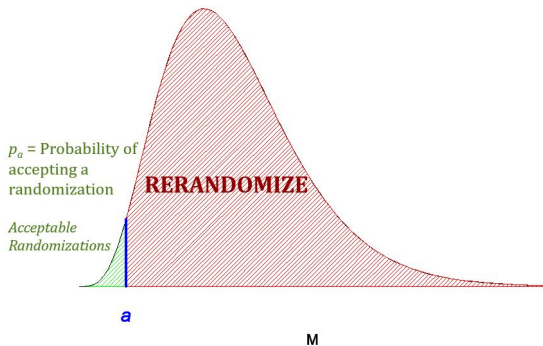
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- A randomization is deemed "acceptable" whenever M falls below a certain threshold, a.
- The rerandomization criterion is :

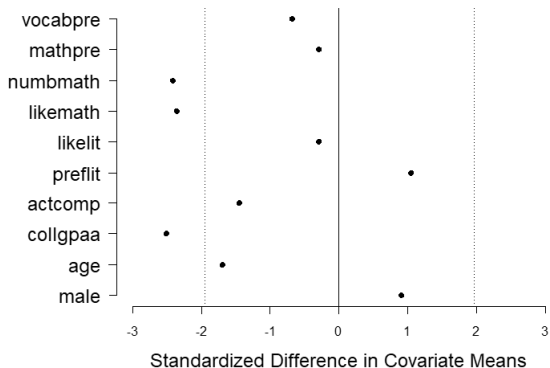
$$\varphi_M(X, W) = \begin{cases} 1 & \text{if } M \leq a \\ 0 & \text{otherwise.} \end{cases}$$

Mahalanobis distance

- Under adequate sample sizes and pure randomization : $M \sim \chi_K^2$



Example : Shadish data



Example : Shadish data

$$\begin{aligned} k &= 10 \text{ covariates} \\ p_a &= 0.001 \end{aligned} \quad P(\chi_{10}^2 \leq a) = 0.001 \Rightarrow a = 1.48$$

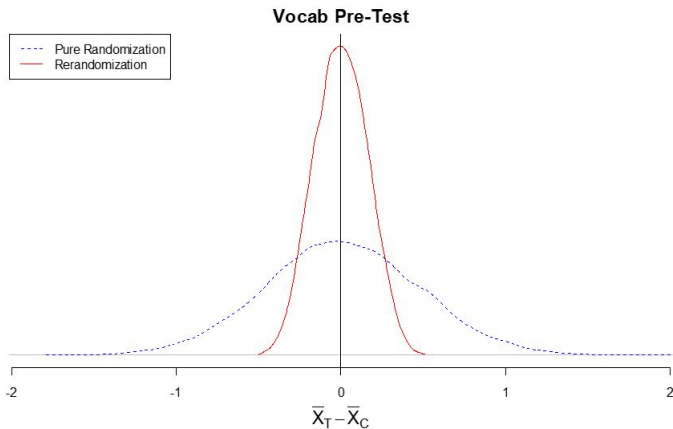
Rerandomize when $M > 1.48$

Ratio of Variances:

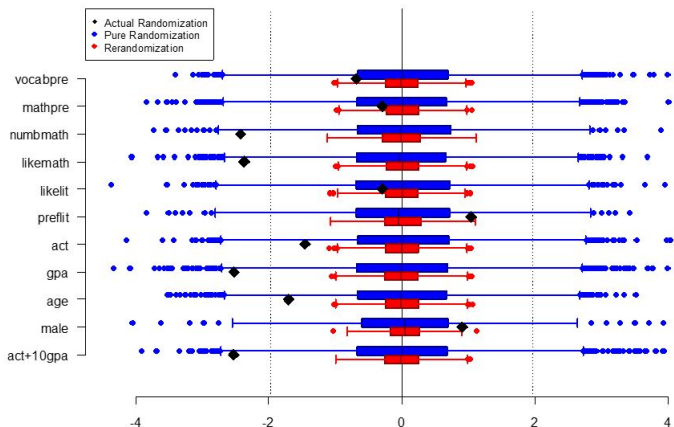
$$v_a = \frac{2}{k} \times \frac{\gamma\left(\frac{k}{2} + 1, \frac{a}{2}\right)}{\gamma\left(\frac{k}{2}, \frac{a}{2}\right)} = \frac{2}{10} \times \frac{\gamma\left(\frac{10}{2} + 1, \frac{1.48}{2}\right)}{\gamma\left(\frac{10}{2}, \frac{1.48}{2}\right)} = 0.12$$

Percent Reduction in Variance: $100(1 - v_a) = 88\%$

Example : Shadish data



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- How to conduct a (Fisher) randomization test?

- Neymanian inference involves a mixture of truncated normals, even asymptotically (Li, Ding, Rubin, 2018).

Conclusions

- Rerandomization improves covariate balance between the treatment groups, giving the researcher more faith that an observed effect is really due to the treatment.
- If the covariates are correlated with the outcome, rerandomization also increases precision in estimating the treatment effect.